Advanced Theory — Module 02: Agents — Study Questions

1. What is a statistical manifold? How does each point on it correspond to a probability distribution?
2. Why is the statistical manifold curved? Give a concrete example using Gaussian parameters.
3. Define the Fisher information matrix. How does it measure the sensitivity of distributions to parameter changes?
4. Show that the Fisher metric is the local quadratic approximation to KL divergence between nearby distributions.
5. Compute the Fisher information matrix for a univariate Gaussian $N(\mu, \sigma^2)$. Interpret the diagonal entries.
6. What is the problem with Euclidean gradient descent on statistical manifolds?
7. Define the natural gradient. How does G^{-1} correct for manifold curvature?
8. Why is the natural gradient parameterization-invariant? Why does this matter?
9. How does natural gradient descent achieve the Cramér-Rao lower bound?
10. How might neural circuits implement the natural gradient? What makes exponential families special?
11. What is meant by "belief updating as geodesic flow"?
12. What are α-connections? Why are the $\alpha = 1$ and $\alpha = -1$ connections important?
13. Explain dual flatness for exponential family distributions. What are natural parameters vs. expectation parameters?
14. What is a Bregman divergence? How does it relate to KL divergence for exponential families?
15. How does the Fisher information matrix change with the precision of a Gaussian?
16. What is the connection between information geometry and Riemannian geometry?
17. How does the geometry of the variational family affect the efficiency of Active Inference?
18. What happens to information geometry when the statistical manifold has boundary points (e.g., $\sigma^2 \to 0$)?
19. How does the Fisher information relate to the precision of neural representations?

20. Critically evaluate: Is information geometry necessary for understanding Active Inference, or is it a mathematical luxury?